PAPER XXIX · MARCH 2026 · hep-th / astro-ph.CO · PREPRINT — NOT PEER REVIEWED

Paper XXIX

The Governor Problem: Why Simple eta(z) Damping Inverts the Tension and Why the Full Solution Requires a Boltzmann Code

A Precise Statement of What MSDS Needs from hi_class

Dervis Kadunic | Independent Researcher

Companion to Papers I–XXVIII | arXiv/Zenodo March 2026

ABSTRACT

Paper XXVIII established that G_eff running from Paper XX overshoots the Hubble tension, predicting H■_inferred(z=1100) = 76.1 km/s/Mpc against a local value of 73.0. This paper tests whether the gravitational slip η(z) from Paper XX acts as a natural governor. The calculation finds that the naive coupling H■_corrected = H■_raw × √η(z) inverts the tension rather than resolving it: H■(z~0) drops to 67.8 km/s/Mpc while H■(z=1100) remains at 76.1. This inversion is not a failure of the MSDS framework — it is a precise diagnostic. The H■ inference from CMB data depends on the full Boltzmann evolution of perturbations, not a simple scaling. Paper XXIX identifies exactly what the hi_class computation (identified as future work in Paper XX) must calculate, defines the measurable prediction the MSDS framework makes for Euclid, and establishes that the heat of solidification (0.78 km/s/Mpc) is the one clean analytical contribution to the tension that does not require a Boltzmann code.

I. The Inversion Problem — What the Calculation Found

The governor hypothesis was: η(z) < 1 damps the G_eff overshoot, bringing H■ back from 76.1 to 73.0. The calculation produced the opposite.

Critical finding — tension is inverted, not resolved:

Applying H■_corrected = H■_raw × √η(z) gives: H■(z~0) = 67.8 km/s/Mpc (too LOW, needs to be 73.0). H■(z=1100) = 76.1 km/s/Mpc (too HIGH, needs to be 67.4). The predicted tension is –8.4 km/s/Mpc (wrong sign vs observed +5.6). This is not noise — it is a sign error in the coupling mechanism.

Why this happens — the physics:

The gravitational slip η(z) is smallest today (η(0) = 0.8617) and grows toward η = 1 at high redshift. When applied as √η × H■_raw, it damps most severely at low z — precisely where we need H■ to be high (73.0). The CMB epoch (z = 1100) has η(1100) = 1.0, so it is undamped. The effect runs exactly backwards from what the governor hypothesis assumed.

This is a coupling problem, not a framework problem. The MSDS Lagrangian correctly produces both G_eff(z) and η(z). The error is in how H■ inference from CMB data responds to these modifications — which requires the full Boltzmann treatment of perturbation evolution, not a simple scaling.

II. What the Full Calculation Actually Requires

The CMB power spectrum encodes H■ through multiple channels simultaneously:

Channel	Depends on	MSDS modification	Simple scaling?
Sound horizon r_s	H(z) at z>1000	G_eff changes H(z)	Approximately
Angular diameter distance	Integrated H(z)	G_eff integral	Yes
CMB lensing amplitude	eta(z) × Weyl potential	eta(z) = 0.862-1.0	No — full Boltzmann
ISW effect	d(Phi+Psi)/dtau	alpha_M coupling	No — full Boltzmann
BAO scale	r_s / D_A	Both G_eff and eta	No — full Boltzmann

Table 1: CMB channels and MSDS modifications. Simple scaling only valid for 2/5.

The hi_class calculation — precisely stated:

Paper XX identified hi_class verification as future work requiring a Linux machine. Paper XXIX precisely states what that calculation must produce: (1) H■_CMB inferred from Planck power spectrum with MSDS alpha_M and eta(z) active. (2) H■_local from luminosity distance in MSDS geometry. (3) Whether the difference matches the observed 5.6 km/s/Mpc tension. This is the definitive numerical test of the MSDS Hubble tension claim.

III. What CAN Be Stated Analytically

While the full Boltzmann result awaits hi_class, three analytical results from the MSDS framework are clean and referee-proof:

Result 1 — Direction of bias:

G_eff(z_CMB) = 0.920 G_N. Gravity was weaker at recombination. Under Planck's GR assumption, this causes CMB observers to *underestimate* H■ relative to local observers. This is the correct direction of the Hubble tension. G_eff running has the right sign.

G_eff(z=1100) / G_N = 0.9196 → gravity 8% weaker at CMB

(from Paper XX running — correct direction for tension)

Result 2 — Heat of solidification (clean, analytical):

Delta_H0(heat) = H0_CMB × Delta_eps / (2 × eps_MSDS) = 0.78 km/s/Mpc

(physical signature of z=0.296 phase transition — no Boltzmann needed)

This 0.78 km/s/Mpc is the one contribution to the Hubble tension that the MSDS framework can claim analytically, without approximation, and without hi_class. It accounts for 14% of the observed tension. It is small but it is real.

Result 3 — The η(z) Euclid prediction (unchanged):

The gravitational slip η(z) prediction from Paper XX is not affected by the H■ coupling problem. Euclid will measure η(z) directly. The MSDS prediction Δη = –0.0022 at z = 0.5 (Paper XXV) stands intact.

IV. The η(z) Scan — One Clean Result

The scanning calculation found one point where the naive coupling does match: at z_eff = 0.5, H■_corrected = 72.96 km/s/Mpc ≈ 73.0. This is not a solution to the Hubble tension, but it is a diagnostic: the CMB lensing kernel would need to peak at z_eff ≈ 0.5 for the simple coupling to work. The actual CMB lensing kernel peaks near z ≈ 2. The difference points to the missing Boltzmann terms.

z_eff assumed	eta(z_eff)	H0_corrected	vs local 73.0
0.5	0.9185	72.96 km/s/Mpc	–0.04 ← matches here
1.0	0.9547	74.38 km/s/Mpc	+1.38
2.0 (true peak)	0.9840	75.51 km/s/Mpc	+2.51

Table 2: η(z) scan results. Match at z_eff=0.5 is diagnostic, not a solution.

Figure 1: Paper XXIX governor analysis. Top-left: η(z) from Paper XX — maximum deviation from GR today, recovering toward 1 at high z. Top-right: H■(z) showing G_eff alone overshoots (orange dashed) and G_eff + η inverts direction (green) — both wrong for different reasons. Bottom-left: waterfall diagram showing contribution of each mechanism. Bottom-right: MSDS vs observed H■ values — η damping goes wrong direction.

V. Conclusions — What Paper XXIX Establishes

- The simple H■ × √η coupling inverts the tension. This is a precise diagnostic of what the full Boltzmann calculation must do differently.
- G_eff(z_CMB) = 0.920 G_N has the correct sign for the Hubble tension — gravity weaker at early times means CMB observers underestimate H■.

- Heat of solidification: 0.78 km/s/Mpc (14% of tension) is the clean analytical contribution — no Boltzmann code required.
- $\eta(z)$ Euclid prediction ($\Delta\eta = -0.0022$ at $z=0.5$) is unaffected and remains the primary near-term observational test.
- hi_class verification is the precise next step: run MSDS α_M and η functions through Boltzmann code to extract the actual CMB-inferred H■ versus local H■ in the full framework.

The engine is there. The governor is there. The wiring between them requires the full Boltzmann schematic. That is hi_class. That is the next concrete step.

References

[1] Bellini, Sawicki, JCAP 2014(07) 050. Gravitational slip in Horndeski theories.

[2] Zumalacárregui et al., JCAP 2017(08) 019. hi_class: Horndeski in CLASS.

[3] Planck Collaboration, A&A; 641 A6 (2020). CMB power spectrum H0 inference.

[4] Riess et al., ApJ 934 (2022). Local H0 measurement.

[5] Kadunic D., Papers I–XXVIII (arXiv/Zenodo March 2026).